Introduction to Robotics and Intelligent Equipment

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Course Description

This course is an introductory course for international undergraduate students majoring in Mechanical Engineering and Electrical Engineering. By systematically analyzing the core knowledge, technology integration logic, and interdisciplinary applications in the fields of mechanical and electrical engineering and automation, it helps students build a professional cognitive framework, stimulate their interest in exploring engineering technology, and clarify their future learning directions.

Instructor:

Dr. Gang Yu (E-mail: gangyu@hit.edu.cn, Office: C326)

Reference materials:

1. Dai Yong, Deng Qianfa et al., Introduction to Mechanical Engineering, Science Press, June 2014;
2. Cui Yujie, Shi Pu, Hua Jianning, Introduction to Mechanical Engineering, Tsinghua University Press, July 2013;
3. Bin Hongzan, ed., Introduction to Mechanical Engineering Discipline, Huazhong University of Science and Technology Press, July 2016;
4. Zhang Xianmin, edited by Chen Zhong, Introduction to Mechanical Engineering, Huazhong University of Science and Technology Press, January 2015;
5. A Brief History of Mechanical Engineering, Tsinghua University Press, July 2015.
6. Jiang Zhenhuan, History of Science and Technology, Shandong Education Press, October 2010.
7. Li Yunjiang, ed., Introduction to Robotics, Mechanical Industry Press, February 2011.
8. Guo Tongying et al., eds., Robotics and Intelligent Control, People's Post and Telecommunications Press, September 2014.

Grade Policy:

Course Project-class quiz (25%)+course report (25%) +course project presentation (50%)

Introduction
-Purpose and function of the course



one Introduction: The purpose and function of the course

- A brief understanding of the development history and prospects of the professional direction of robotics and intelligent equipment
- Preliminary understanding of typical mechanical equipment and manufacturing process
- A preliminary understanding of the basic knowledge system of robotics and intelligent equipment
- Preliminary understanding of robotics and intelligent equipment professional learning methods
- Preliminary understanding of the basic requirements for becoming a robotics and intelligent equipment professional
- Outline the professional knowledge system of relevant professional direction, and provide guidance for professional learning
- Cultivate students 'professional interests and provide guidance for students' career choices



two

Development of Robotics and Intelligent Equipment Professional Technology

1. Several basic concepts

2. A Brief History of Mechanical Design

3. A Brief History of Mechanical Manufacturing

4. Development history of mechanical and electronic engineering

5. Formation and Development of Robot Engineering

The development course of the relevant direction of this professional category-several basic concepts

1) Machinery（机械） and machine (机器)

- "机" originally refers to a local key component;
- "机械" in ancient China originally refers to a whole instrument or apparatus;
- Mechine: "Ingenious Design"
- Machinery refers to the general term for machines and mechanisms;
- Machinery is a device composed of components connected in different ways and having certain motion relations, which is used to help people reduce the difficulty of work, save people's physical strength, extend people's abilities or realize functions that people cannot.

The development course of the relevant direction of this professional category-several basic concepts

The main characteristics of machinery are manifested in three aspects:

1) A machinery is a combination of artificial physical components;

2) There is a certain relative motion between the various parts of the machinery.

3) can complete the energy, motion transmission and transformation









How do wind turbines work
https://b23.tv/cWFYgwH

two The development process of the relevant directions of this professional category-several basic concepts

1) Machinery and machine

- **Machine:** a device or equipment that has a certain relative motion relationship between parts and components and can complete the transmission and transformation of energy and motion.

- **The machine is** generally composed of support, power, transmission, execution, manipulation, control and other parts, among which:

support portion power section transmission part

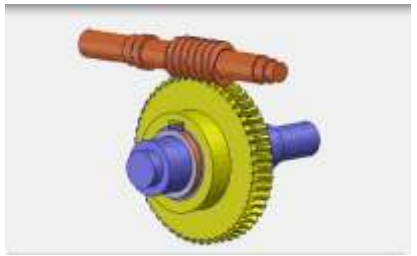
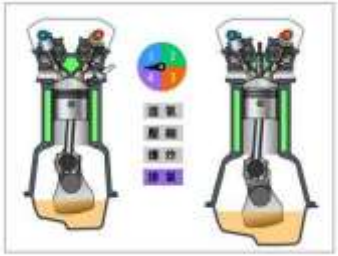
manipulating portion control segment execution part

- Generalization of the concept of "machine": are mobile phones and TV sets machines?

- **Mechanism:** A combination of two or more components that are movably coupled to achieve a specified motion.







The development course of the relevant direction of this professional category-several basic concepts

2)Machining and manufaturing

- **Machining/Processing:** **The process of changing the shape size or performance of a workpiece by mechanical equipment.**

  Such as: Cutting machining Grinding machining Special machining

- **Mechanical manufacturing (manufacturing):** **The methods, techniques and processes to achieve parts processing, product assembly, quality control and process management of mechanical products.**







The development course of the relevant direction of this professional category-several basic concepts

3)Engineering and Mechanical Engineering

- **Engineering**: Based on an envisaged goal and application of relevant scientific knowledge, technical means, and practical experience, organizing a group of people to transform an existing entity (or entities)(natural or man-made) into a man-made product with expected use value.

(1) Research: Apply mathematical and natural science concepts, principles, experimental techniques, etc. to explore new working principles and methods.

(2) Development: Solve various problems encountered in applying research results to practical processes.

(3) Design: Select different methods, specific materials and determine design solutions that meet technical requirements and performance specifications.

(4) Construction: Preparation of site, material storage, selection of work steps, organization of personnel and utilization of equipment.

(5) Production: In consideration of human and economic factors, select factory layout, production equipment, tools, materials, components and process flow, and conduct product testing and inspection.

The development course of the relevant direction of this professional category-several basic concepts



3)Engineering and Mechanical Engineering

(6) Operation: Manage machines, equipment and power supply, transportation and communication, so that all kinds of equipment can operate economically and reliably.

(7) Management and other functions.

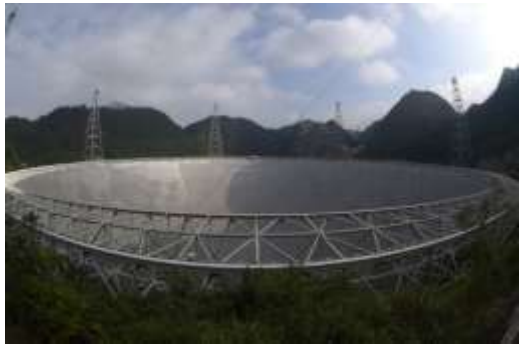
- **Engineering is an applied discipline, a field of knowledge of methods, techniques, and processes that explores and applies the principles of mathematics and other natural sciences to design and make useful objects. It mainly studies the application mode and method of natural science as well as the general law of engineering implementation and improvement applied in various industries.**

- **Those who practice engineering are called engineers.**









The development course of the relevant direction of this professional category-several basic concepts

3)Engineering and Mechanical Engineering

- **Mechanical Engineering:** the activities and processes of applying scientific knowledge, technical means and practical experience to carry out research and development, design, manufacture, installation, application and maintenance of mechanical products and equipment and related problems in a purposeful, systematic and organized manner.
- **Mechanical Engineering:** It is also the basic unit of knowledge classification and talent training, namely the so-called disciplines and majors, mainly covering three academic branches of mechanical design, mechanical manufacturing and mechanical and electronic engineering (electromechanical automation) and several branches of industrial fields such as vehicle engineering, robot engineering and textile machinery.
- **Mechanical Engineering** was the main technical factor contributing to the industrial revolution and capitalist mechanical mass production in the 18th and 19th centuries.

two The development history of the relevant direction of this professional category-A brief history of the development of mechanical design

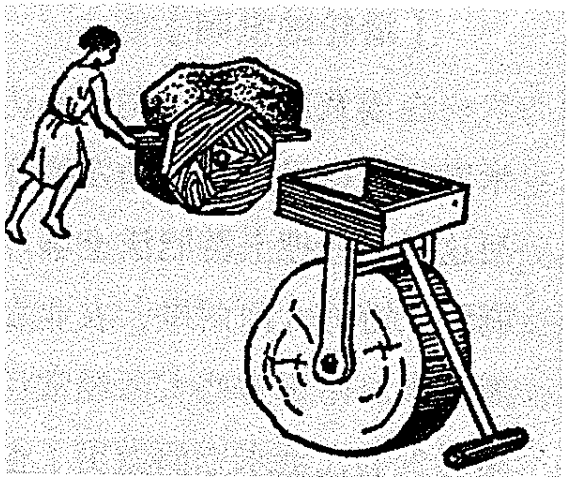
1) The first stage of mechanical design development (pre-Renaissance)

(1) Main contributions:

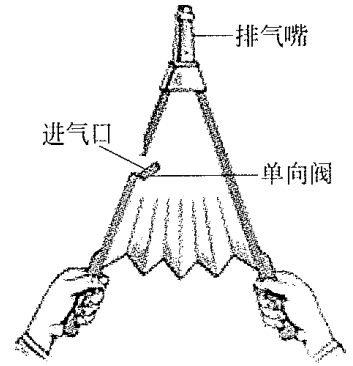
Found: cylinder and lever have labor-saving function; The special function of triangular inclined plane; Energy storage function of elastic component;

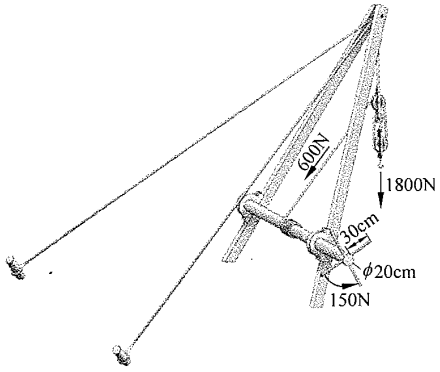
Realized: simple mechanism application to multi-mechanism combination application; manpower, energy storage to natural power machinery.

(2) Use of materials: stone (wood) age, bronze age, iron Age.

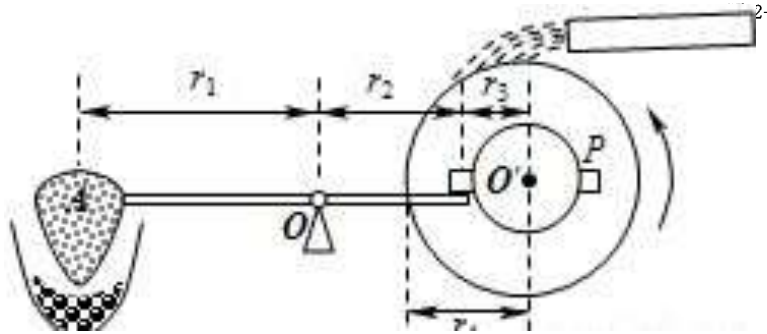








2-17 古希腊时期的起重机



two The development history of the relevant direction of this professional category-

A brief history of the development of mechanical design

1) The first stage of mechanical design development (pre-Renaissance)

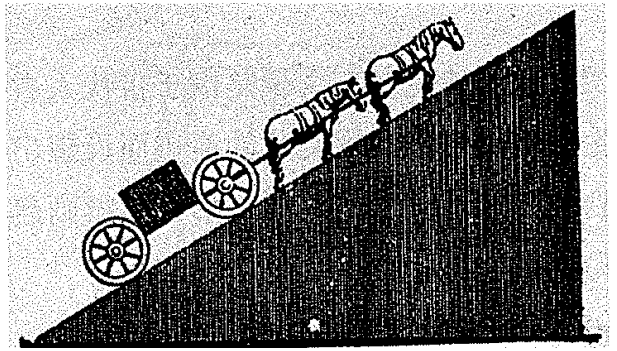
(1) Main contributions:

Found: cylinder and lever have labor-saving function; The special function of triangular inclined plane; Energy storage function of elastic component;

Realized: simple mechanism application to multi-mechanism combination application; manpower, energy storage to natural power machinery.

(2) Use of materials: stone (wood) age, bronze age, iron Age.





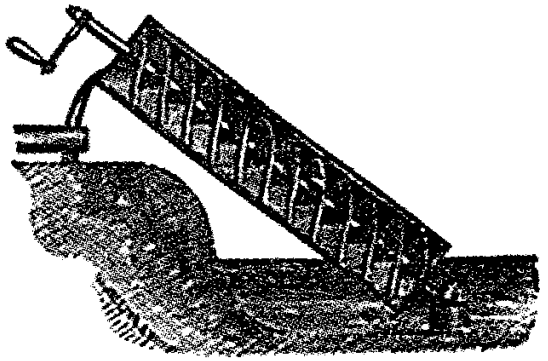


图 2-18　阿基米德的螺旋输水机

two The development history of the relevant direction of this professional category-

A brief history of the development of mechanical design

1) The first stage of mechanical design development (pre-Renaissance)

(1) Main contributions:

Found: cylinder and lever have labor-saving function; The special function of triangular inclined plane; Energy storage function of elastic component;

Realized: simple mechanism application to multi-mechanism combination application; manpower, energy storage to natural power machinery.

(2) Use of materials:stone (wood) age, bronze age, iron Age.

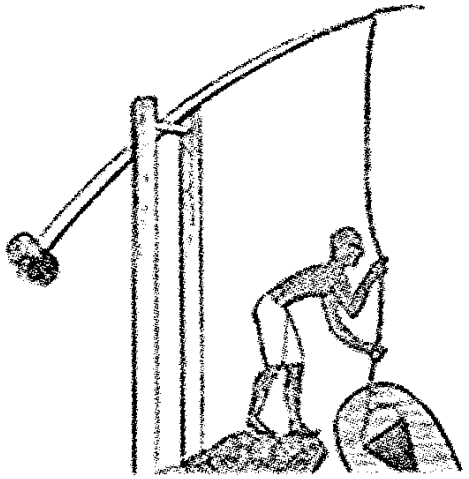
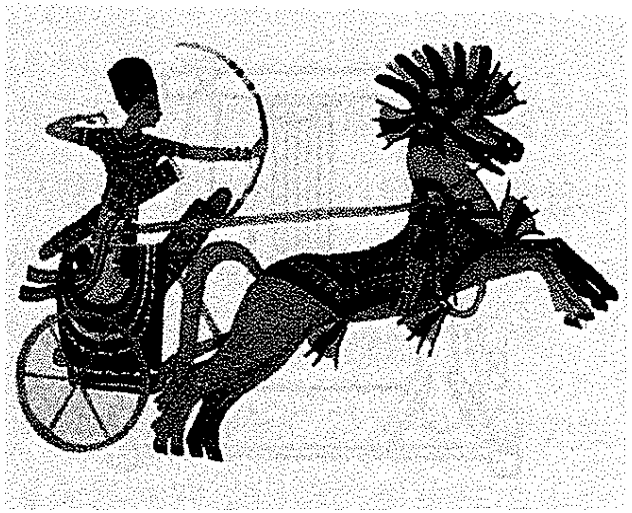


图 2-3　古埃及的桔槔取水



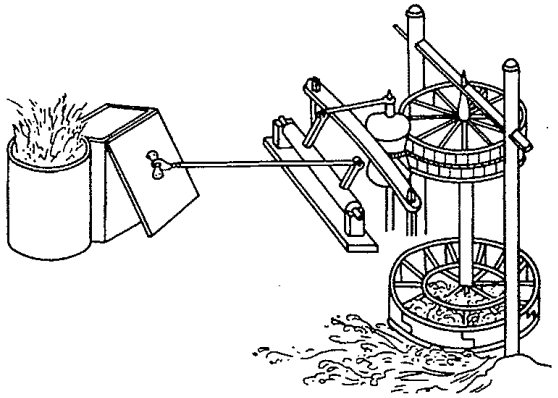


图 2-8　东汉时期的鼓风机——水排

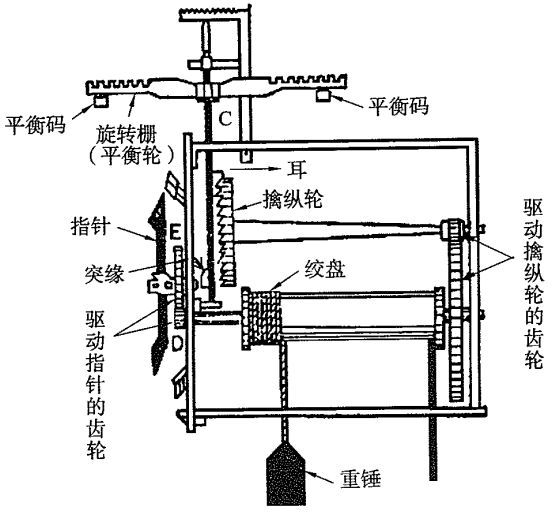


图2-30　德维克的时钟

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The development history of the relevant direction of this professional category-

A brief history of the development of mechanical design

1) The first stage of mechanical design development (pre-Renaissance)



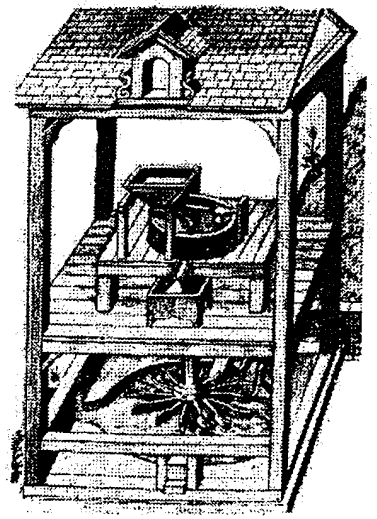


图2-21 挪威型水车

(1) Main contributions:

Found: cylinder and lever have labor-saving function; The special function of triangular inclined plane; Energy storage function of elastic component;

Realized: simple mechanism application to multi-mechanism combination application; manpower, energy storage to natural power machinery.



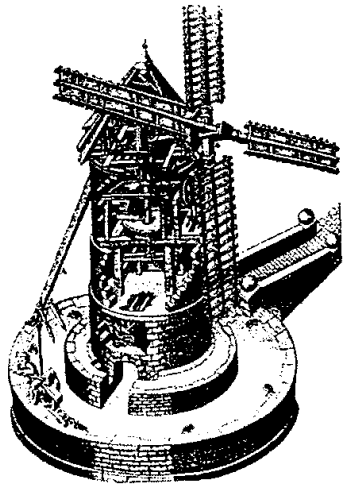


图2-27 塔型风车

(2) Use of materials: stone (wood) age, bronze age, iron Age.

two The development history of the relevant direction of this professional category-A brief history of the development of mechanical design

1) The first stage of mechanical design development (pre-Renaissance)

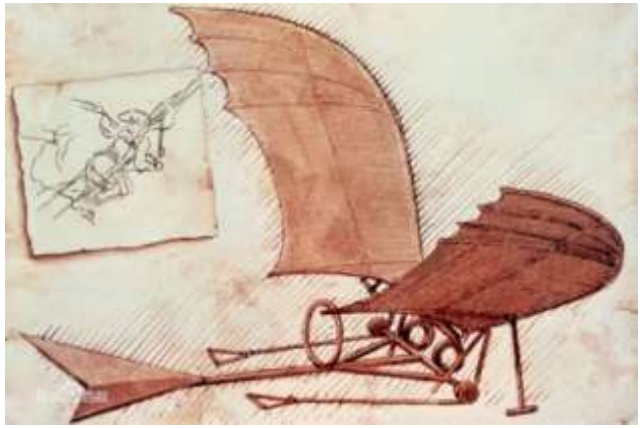
(3) Summary of achievements and characteristics:

- **With the transmission mechanism as the main line, a simple machinery characterized by lever, pulley, wheel shaft, inclined plane, spiral and wedge has been formed, which has laid the foundation for mechanical development, and the focus of development is on mechanical transmission.**
- **Mathematics and mechanics are not developed enough, and innovative design depends on intuition, inspiration and experience. Archimedes is an outstanding representative of the rational development of machinery.**
- **Material: stone, wood gradually developed to copper, iron.**
- **Power is mainly manpower and stored power, gradually realizing, recognizing and utilizing natural power (wind power, hydraulic power) and artificial power (potential energy, steam power)**
- **On Needham (李约瑟) Question: "Why didn't scientific breakthroughs and industrial revolution happen in China?"**

two The development history of the relevant direction of this professional category-A brief history of the development of mechanical design

2) The second stage of mechanical design development (Renaissance to World War II)





(1) Main contributions:

■ **The breakthrough and rapid development of mathematics, physics and mechanics promoted the construction of the theoretical basis of mechanical design.**

Leonardo. Da. Finch: Artist and inventor of many mechanical devices: lathes, boring machines, threading machines, continuously variable transmissions. Concept ideas: helicopters, machine guns, tanks, robots, calculators, etc.





Galileo Galilei: mathematician, astronomer, mechanic. Velocity, acceleration, parabolic motion and free fall motion are defined, and inertia theorem is deduced.

Descartes: defined rectangular coordinate system, established analytical geometry, momentum theorem.

The development history of the relevant direction of this professional category-A brief history of the development of mechanical design

2) The second stage of mechanical design development (Renaissance to World War II)

(1) Main contributions:

■ The breakthrough and rapid development of mathematics, physics and mechanics promoted the construction of the theoretical basis of mechanical design.





Isaac Newton (1642–1727)

Huygens: momentum theorem, simple pendulum motion, concept of centrifugal force, centripetal acceleration in relation to circumferential velocity and radius





Newton: mathematician, astronomer, physicist, binomial theorem, founding differential geometry, law of universal gravitation, Newton's three laws, mathematical principles of natural philosophy.

D'Alembert: French mathematician and dynamicist. Research on constrained object motion, D'Alembert principle, mechanical dynamics analysis.

Euler: scientific generalist, Euler angle, rigid body motion around a point, moment of inertia, laying the foundation of rigid body dynamics

The development history of the relevant direction of this professional category-A brief history of the development of mechanical design

2) The second stage of mechanical design development (Renaissance to World War II)

(1) Main contributions:

■ **The breakthrough and rapid development of mathematics, physics, and mechanics has promoted the construction of the theoretical basis of mechanical design;**





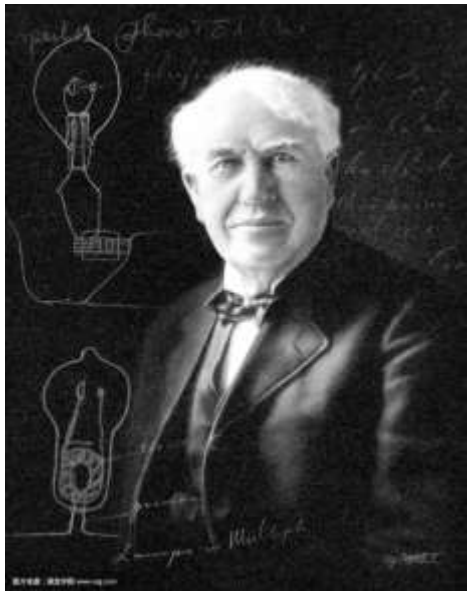
Faraday: Discovering the Principle of Electromagnetic Induction



Carnot: theoretical calculation of Carnot cycle and heat engine

Maxwell: established a complete theory of electromagnetic fields

Wiener. Siemens: Base self-excited DC generator

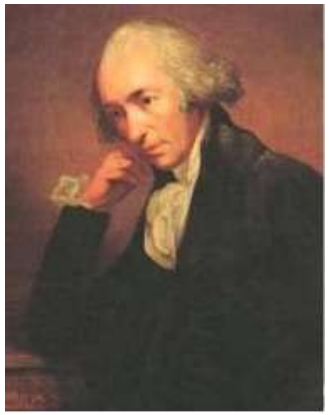


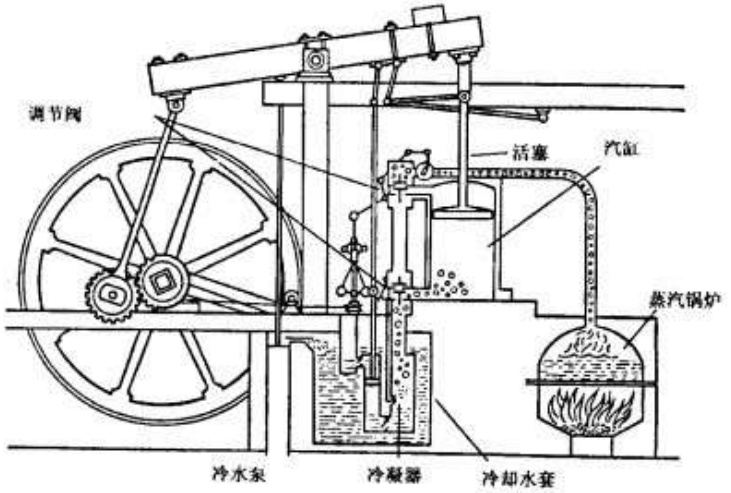
Edison: Direct current power station.

The development history of the relevant direction of this professional category-
A brief history of the development of mechanical design

2) The second stage of mechanical design development

(Renaissance to World War II)



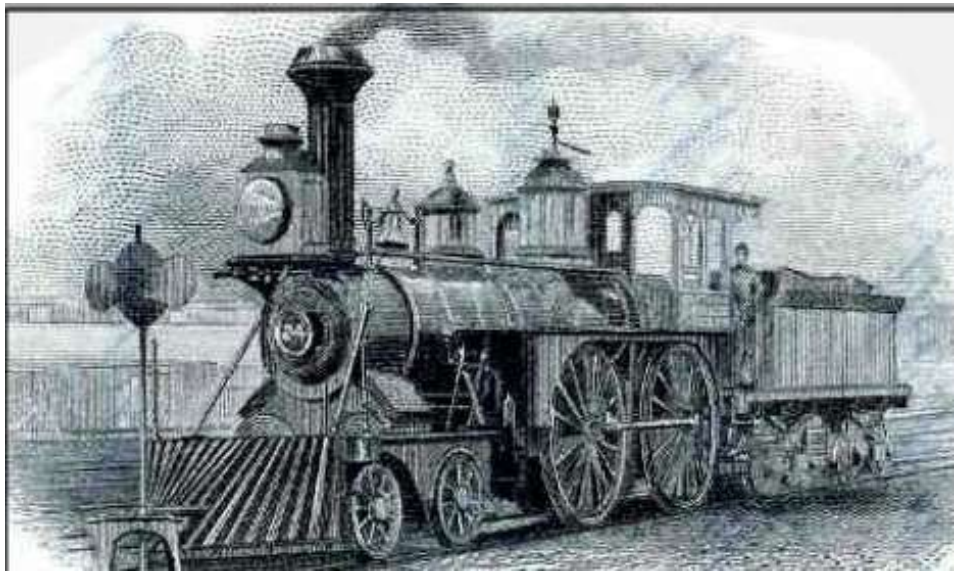


(1) Main contributions:

- **Watt's improvement of steam engine and Carnot's theorem gave birth to the rapid development of power machinery;**

Watt steam engine: **1) single condenser; 2) rotary motion output; 3) double-acting cylinder; 4) flywheel speed regulation; 5) centrifugal governor.**

Produced: **Trevitz steam engine: 1804, 15 tons, 8km/h. Stevenson steam engine: 1814, 21 knots, 15km/h.**



The development history of the relevant direction of this professional category-
A brief history of the development of mechanical design

2) The second stage of mechanical design development (Renaissance to World War II)

(1) Main contributions:

■ **Watt's improvement of steam engine and Carnot's theorem gave birth to the rapid development of power machinery;**

Internal combustion engine: 1862, invented by Durocher, 80-150r/min, efficiency 12%-14%, power density 272kg/kW

Alto, Daimler, Benz, Ford, a large number of automotive applications.

Steam turbine: 1854, 10 horsepower multistage reaction steam turbine;

Hydraulic turbine: Francis reaction turbine, 1850;

Pelton impulse turbine, 1889;

axial flow turbine, 1913.

Gas turbine jet engine of 1921

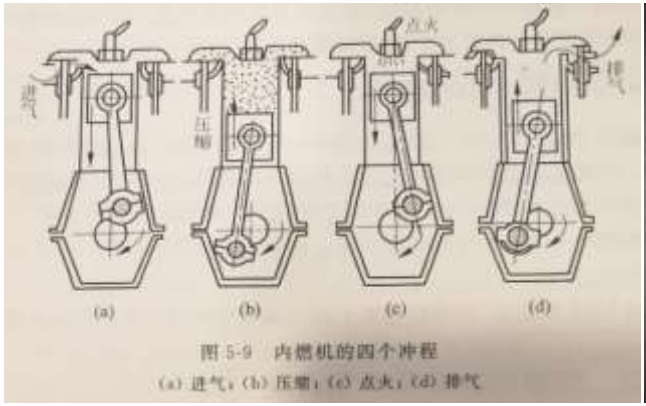
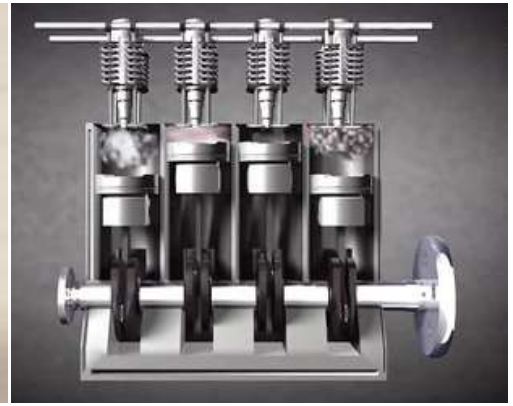


图 5-9 内燃机的四个冲程
(a) 进气; (b) 压缩; (c) 点火; (d) 排气



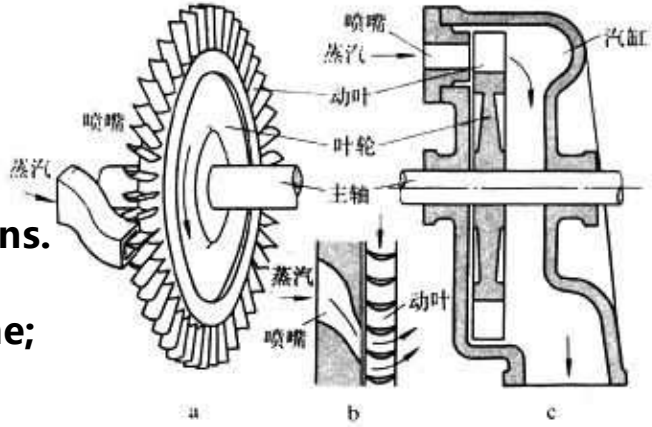
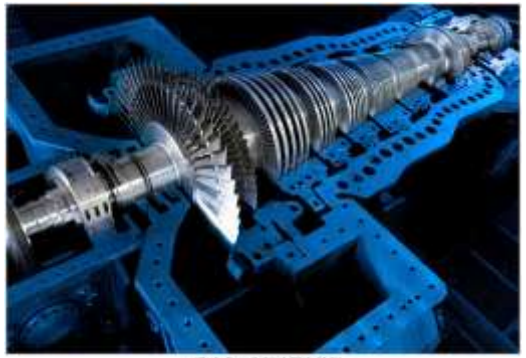
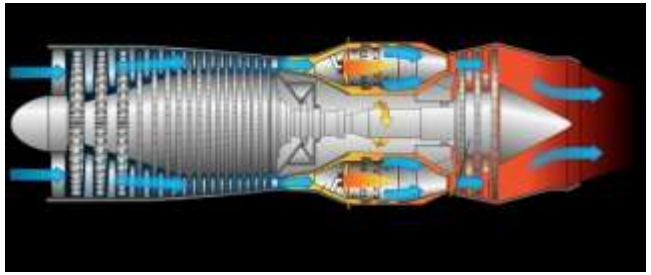


图 1 单级汽轮机


反动式工业驱动用汽轮机



The development history of the relevant direction of this professional category-
A brief history of the development of mechanical design

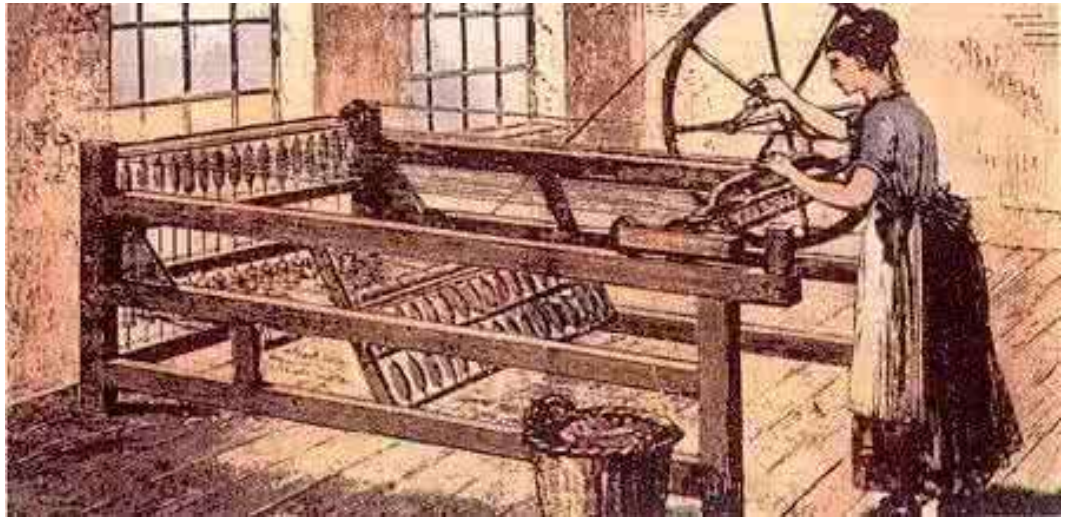
2) The second stage of mechanical design development (Renaissance to World War II)





(1) Main contributions:

- **The invention of Jenny spinning machine and sewing machine accelerated the development of mechanism and formed the rudiment of mass production mode.**



Jenny Loom: Invented by James Hargreaves in 1764; the advent of the Jenny Loom enabled the establishment of large-scale looms. The invention of the engine was the beginning of the first industrial revolution. The spinning capacity of the Jenny machine is eight times higher than that of the old spinning wheel

Sewing machine Timenier sewing machine, 1830

The development history of the relevant direction of this professional category-A brief history of the development of mechanical design

2) The second stage of mechanical design development (Renaissance to World War II)

(2) Brief summary:

- With power machinery as the focus, the power machinery system was formed, and the transformation from natural power to artificial power system was realized. Two industrial revolutions, entered the electric power era.
- The development of textile machinery, mining machinery and transportation machinery has accelerated the development of mechanism science and formed the design system of transmission mechanism and executive mechanism.
- Hydraulic, agricultural machinery development, gave birth to the development of fluid machinery,
- With the development of mathematics and dynamics, mechanical design relying on intuition, inspiration and experience has developed into rational design relying on theory, forming the basic theoretical system of mechanical design.
- Material: copper, iron gradually developed to steel.

two The development history of the relevant direction of this professional category-A brief history of the development of mechanical design

3) The third stage of mechanical design development (after World War II)

- **Changes in development needs:** war and national security, technology-led economic development, increasingly prominent resources (including energy) and environmental issues, unprecedented attention to humanized design, and the expansion of human activity space to both poles
- **Major theoretical and technological breakthroughs:** physics, biology, computer-centered information technology, perception technology, artificial intelligence have made major breakthroughs, new materials emerge one after another.

(1) Main contributions:

■ **With the development of mathematics, physics, dynamics and manufacturing technology, the theoretical system of mechanical design has been deepened and improved**

- Mechanism (mechanism kinematics, emerging mechanism, mechanism dynamics)
- Mechanical transmission (new principle, stepless transmission)
- Mechanical dynamics (analysis, modeling, simulation)
- Mechanical strength science (fatigue design theory, fracture mechanics)
- tribological
- Robotics (robotics, robotics dynamics)
- Microelectromechanical Systems (MEMS) Design Theory
- hydraulic transmission and control

The development history of the relevant direction of this professional category-
A brief history of the development of mechanical design

3) The third stage of mechanical design development (after World War II)

(1) Main contributions:

- **With the development of computer science and technology, mechanical design methods and means have achieved fundamental changes.**

- **In 1963, MIT's I. Sutherland completed the design of graphical interactions between humans and computers.**

- **1970s CAD (Computer-Aided Design) was widely used. Since the 1980s, numerical simulation, integrated analysis, optimization design, virtual reality (VR) and other increasingly mature and large-scale applications. AutoCAD, Solidworks, CATTIA, Pro/Engineer and other software systems are becoming more and more powerful, and the era of hand calculation and hand drawing is gone forever.**

- **The development and application of digital twins and artificial intelligence provide new opportunities and challenges for mechanical design and optimization and simulation.**

- **Mechanical design methodology develops in depth with the development of electromechanical technology, big data and artificial intelligence**

  Optimization design, concurrent engineering, reverse engineering, reliability design

two The development history of the relevant direction of this professional category-A brief history of the development of mechanical design

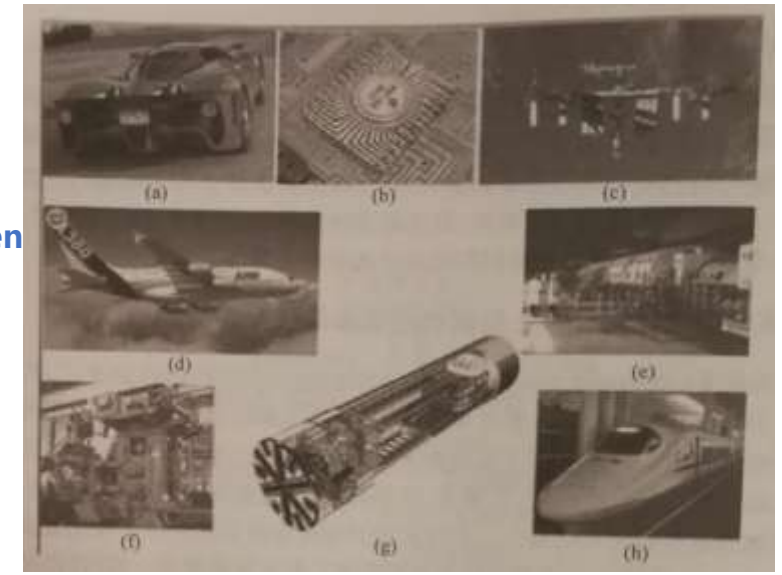
3) The third stage of mechanical design development (after World War II)

(1) Main contributions:

- **In response to the development of high-tech technology and industry, new breakthroughs have been made in mechanical system design.**

Large-scale, miniaturization, precision, complexity, personalization, high reliability, environmental coordination, life-cycle optimization and other requirements have become the content that needs to be fully considered in mechanical design, not just the function and performance of the product.

- **A380 power: 230MW, high-speed train speed 574.8km/h.**
- **The capacity of steam turbine unit reaches 1500MW.**
- **German Krupp-288 excavator 95m high, 215m long, digging diameter 21m.**
- **Electromechanical systems represented by bullet trains, high-speed trains, launch vehicles, aircraft carriers, etc. have achieved a high degree of integration of "machinery, electricity, fluid and light."**





3) The third stage of mechanical design development

(after World War II)

(2) Brief summary:

During this period, the vision of design was greatly expanded, the design theory was constantly improved, the connotation of design was constantly enriched, and the design means made a revolutionary breakthrough, which was the most brilliant period in history.

two

The development history of the relevant direction of this professional category-A brief history of the development of mechanical manufacturing

1) The first stage of mechanical manufacturing development (before the Renaissance)

(1) Main contributions: the rudiments of manufacturing technology and equipment appeared

- **Cutting Technology and Equipment**
- **Foundry technology: 6000 BC sand casting, China 21-17 BC, bronze casting**
- **Forging technology: cutting, forging, bending, annealing, grinding process**

(2) Use of materials: stone (wood) age, bronze age, Iron Age.



图 2-25　古埃及加工木制工件的车床



图 2-26　13 世纪欧洲的车床



23　《天工开物》中描述的锻制千钧锚的情况

Development history of related directions in this major-A brief history of the development of mechanical manufacturing

1) The first stage of mechanical manufacturing development (before the Renaissance)

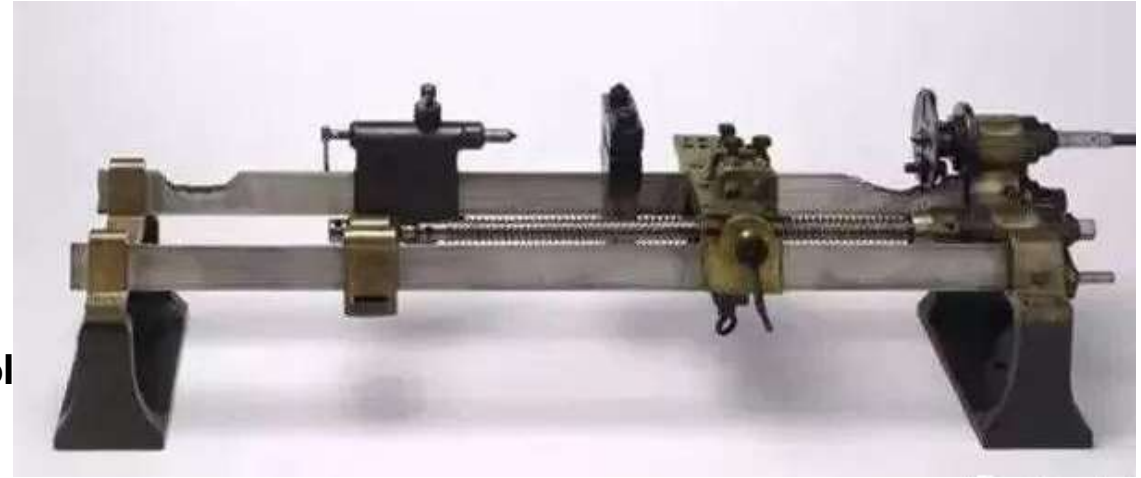
(3) Brief summary: the prototype of manufacturing technology and equipment appeared

- **Based on manpower, the basic technological process of machining mechanical parts and the basic form of equipment have been formed.**
- **The manufacturing level depends on human skills, the accuracy of equipment is weak, and the production efficiency is low.**
- **Material: stone, wood gradually developed to copper, iron.**

The development history of the relevant direction of this professional category-
A brief history of the development of mechanical manufacturing

2) The second stage of mechanical manufacturing development

(Renaissance to World War II)



(1) Main contributions:

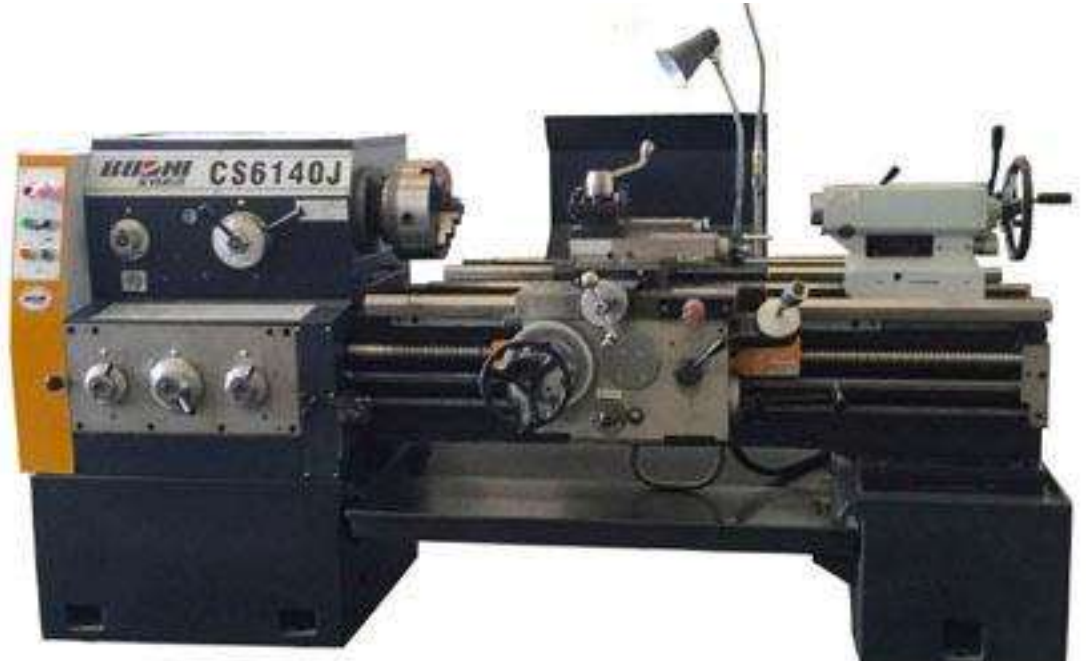
■ The basic form of the machine tool is constructed.

Modern concept of machine tool: machining metal, machine traction tool holder movement

Invention of boring machine: Wilkinson, England, 1774

Manufacturing steam engine cylinder bore: 1.27m diameter, error 1.6mm.

19th century: During the rapid development of machine tools, electric motors replaced steam engines, and automatic lathes, grinders, gear processing machines, etc. were invented successively. Milling machines and grinders were greatly improved.



Development history of related directions in this major-A brief history of the development of mechanical manufacturing

2) The second stage of mechanical manufacturing development (Renaissance to World War II)

(1) Main contributions:

■Important progress has been made in the development of tool materials;

- In 1868, R. Mushet invented tool steel containing tungsten alloy, and the cutting speed of the tool reached 18m/min.
- 1890s, American F. Taylor invented high-speed steel, cutting speed up to 36.5m/min.
- In 1893, H. Moissan discovered tungsten carbide with cutting speeds of more than 100m/min.

■Emergence of interchangeable production

- In 1798, American Whitney contracted for 15000 guns.
- In 1801, guns were produced in an interchangeable manner.
- In 1841, standardized threads of uniform size were established to establish standards for interchangeability.

The development history of the relevant direction of this professional category-
A brief history of the development of mechanical manufacturing

2) The second stage of mechanical manufacturing development (Renaissance to World War II)

(1) Main contributions:

- F. Taylor's Scientific Management System
- Ford's first mass production model and assembly line
- Standardization and serialization.

(2) Brief summary:

- **Changing the drive of the machine tool to a motor opens up room for improvement of the machine tool.**
- **The development of tool materials plays a crucial role.**
- **The production organization has undergone fundamental changes and entered the era of scientific management.**
- **Interchangeability laid the foundation for mass production, the design concept of the production process changed significantly, the rhythm production mode was born, and a new era of the assembly line was initiated.**
- **Standardized, serialized, rigid production lines provide a broad space for reducing production costs.**

two Development history of related directions in this major-A brief history of the development of mechanical manufacturing

3) The third stage of mechanical manufacturing development (after World War II)

- **Changes in development needs:** war and national security, technology-led economic development, increasingly prominent resources (including energy) and environmental issues, unprecedented attention to humanized design, and the expansion of human activity space to both poles
- **Major theoretical and technological breakthroughs:** physics, biology, computer-centered information technology, perception technology, artificial intelligence have made major breakthroughs, new materials emerge one after another.

(1) Main contributions:

- **In addition to turning, milling, grinding, drilling, casting, forging, welding parts processing methods, a large number of new processing methods have emerged, enriching and perfecting the machining method system, theoretical system and technical system.**

EDM, wire cutting, electron beam machining, ion beam machining, laser machining, water jet machining, etc.








two

The development history of the relevant direction of this professional category-

A brief history of the development of mechanical manufacturing

3) The third stage of mechanical manufacturing development (after World War II)

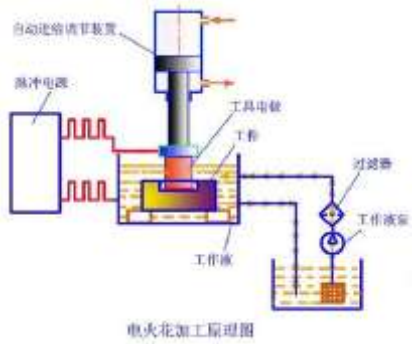
(1) Main contributions:

- **In addition to turning, milling, grinding, drilling, casting, forging, welding parts processing methods, a large number of new processing methods have emerged, enriching and perfecting the machining method system, theoretical system and technical system.**

EDM, wire cutting, electron beam machining, ion beam machining, laser machining, water jet machining, powder metallurgy, etc.







电火花加工原理图

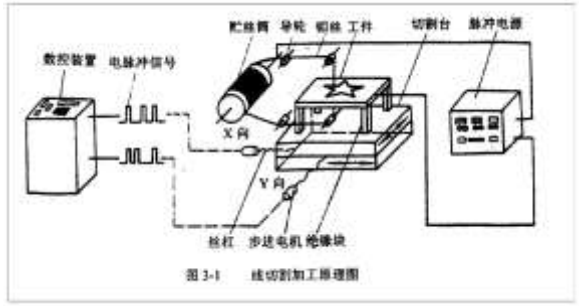


图 3-1 线切割加工原理图

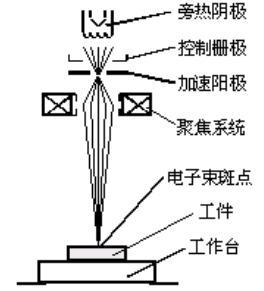
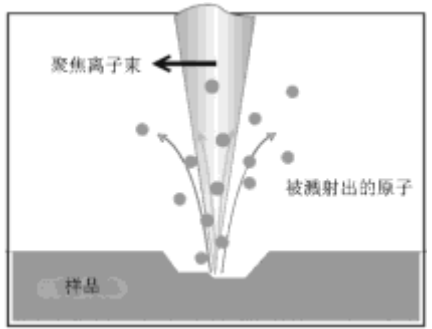
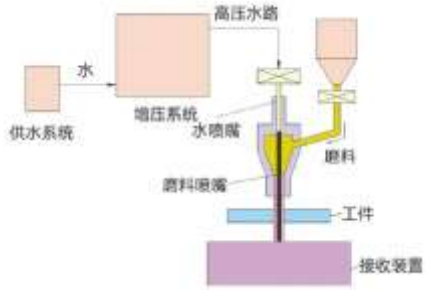


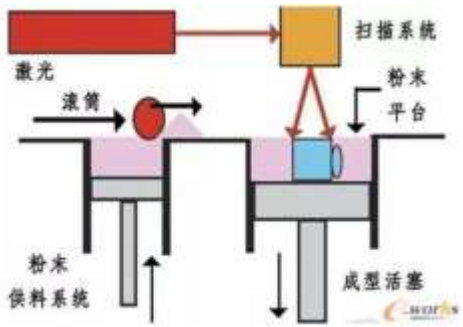
图 7-20 电子束加工原理图









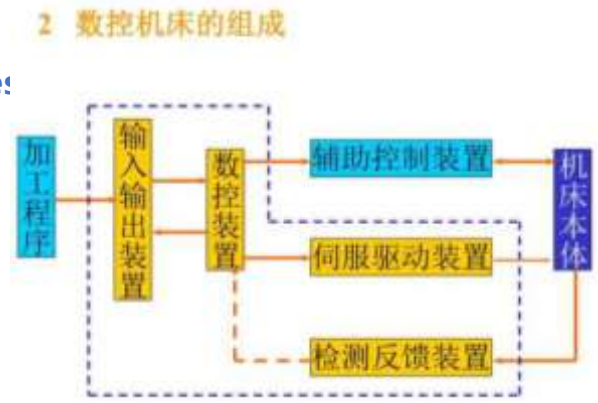


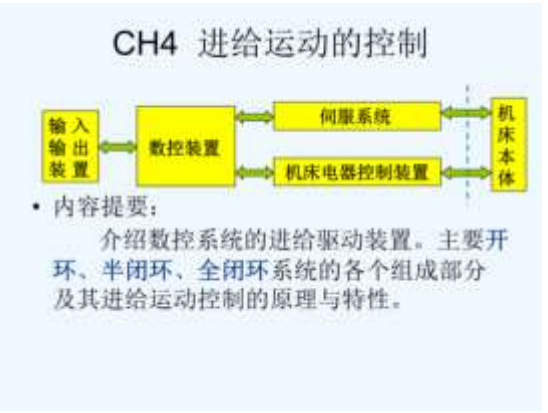
The development history of the relevant direction of this professional category-
A brief history of the development of mechanical manufacturing

3) The third stage of mechanical manufacturing development (after World War II)

(1) Main contributions:

- **The development of numerical control technology of machine tools promotes the popularization and application of numerical control technology of electromechanical equipment, and also promotes the transformation of part machining process design concept from process dispersion to process concentration**





- **MIT's computer-controlled machine tool in 1952**
- **1952-1958 MIT's APT language, CAM;**
- **1958 Machining center;**
- **1966 Computer group control system;**
- **1970s Flexible Manufacturing Unit.**
- **Intelligent Manufacturing Cell after 2000**





The development history of the relevant direction of this professional category-A brief history of the development of mechanical manufacturing

3) The third stage of mechanical manufacturing development (after World War II)

(1) Main contributions:

- **With the development of computer science and technology, mechanical design methods and means have achieved fundamental changes.**

- **1950s, Group Technology**
- **Computer-aided process planning (CAPP).**
- **1970s: CAPP/CAM is widely used. Since the 1980s, numerical simulation, integrated analysis, optimization design, virtual reality (VR) and other increasingly mature and large-scale applications. Solidworks, CATTIA, Pro/Engineer and other software systems are becoming more and more powerful, and the era of manual calculation and manual drawing is gone forever.**
- **Integration of CAD/CAM/CAPP**

The development history of the relevant direction of this professional category-
A brief history of the development of mechanical manufacturing

3) The third stage of mechanical manufacturing development (after World War II)

(1) Main contributions:

- **Breakthroughs in high-speed machining technology have greatly improved cutting efficiency and promoted the development of a series of related technologies.**

- **High speed cutting concept: C. Salomon,**
- **Technological breakthrough: around 2000, the cutting speed is now 1500m/min**
- **High-speed machine tools, high-speed CNC, high-speed tools, etc.**
- **Compared with traditional machining, high speed machining has undergone an essential leap, the metal removal rate per unit power has increased by 30%~40%, the cutting force has decreased by 30%, and the cutting life of the tool has increased by 70%.**

- **Precision and ultra-precision manufacturing technology**

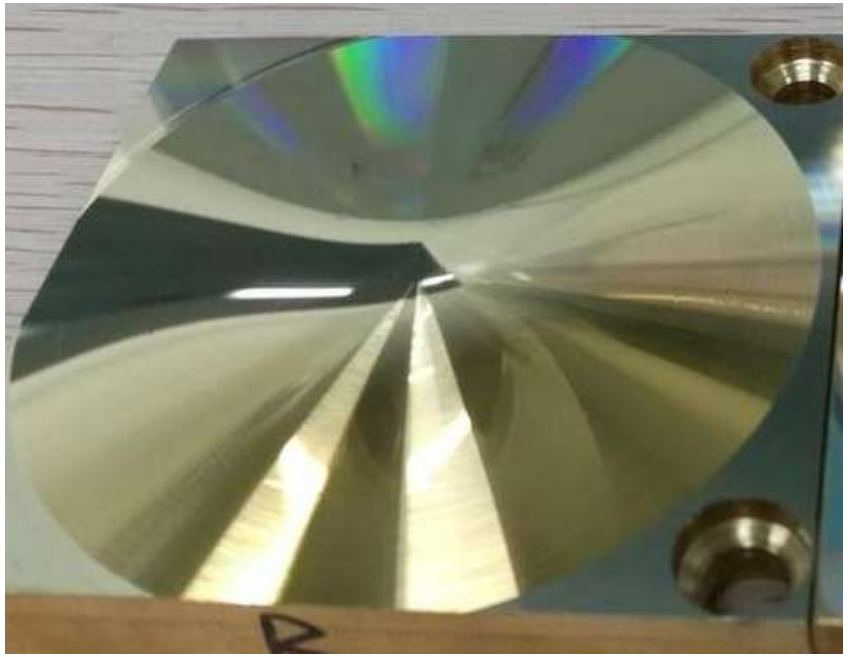
The development history of the relevant direction of this professional category-A brief history of the development of mechanical manufacturing

3) The third stage of mechanical manufacturing development (after World War II)

(1) Main contributions:

■ **Precision and ultra-precision manufacturing technology**

- **1960s, adapted to the needs of cutting-edge technologies such as nuclear energy, large-scale integrated circuits, lasers and aerospace;**
- **Process methods: ultra-precision turning, mirror grinding and grinding, electrochemical machining, mechanical-chemical polishing, ion sputtering and ion implantation, electron beam exposure, laser beam machining, and metal evaporation.**
- **Accuracy: The highest accuracy can reach 0.1 micron, and the surface roughness is Rz 0.05 micron.**





Development history of related directions in this major-A brief history of the development of mechanical manufacturing

3) The third stage of mechanical manufacturing development (after World War II)

(1) Main contributions:



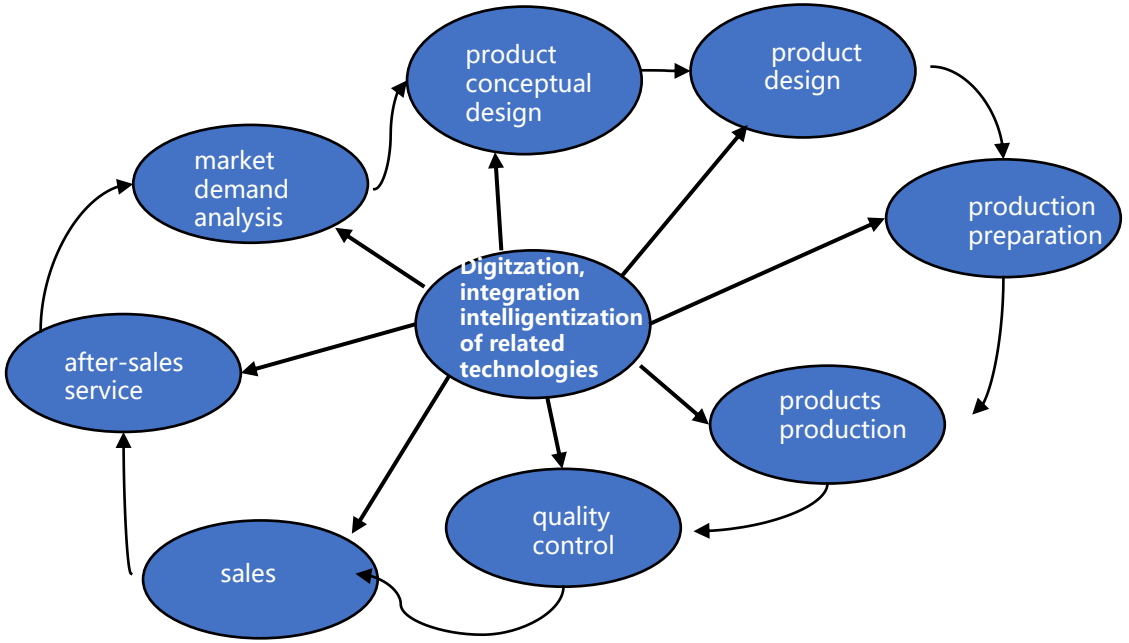
- **Unprecedented Development of Cutting Tool Technology**

Tool materials: carbide, coating technology, ceramic materials

Breakthrough progress in tool manufacturing technology and tool use methods

- **Flexible manufacturing technology and computer integrated manufacturing system have promoted significant changes in processing technology and production organization management.**

- **The application of artificial intelligence provides a broad application space for intelligent manufacturing**



Development history of related directions in this major-A brief history of the development of mechanical manufacturing

3) The third stage of mechanical manufacturing development (after World War II)

(2) Brief summary:

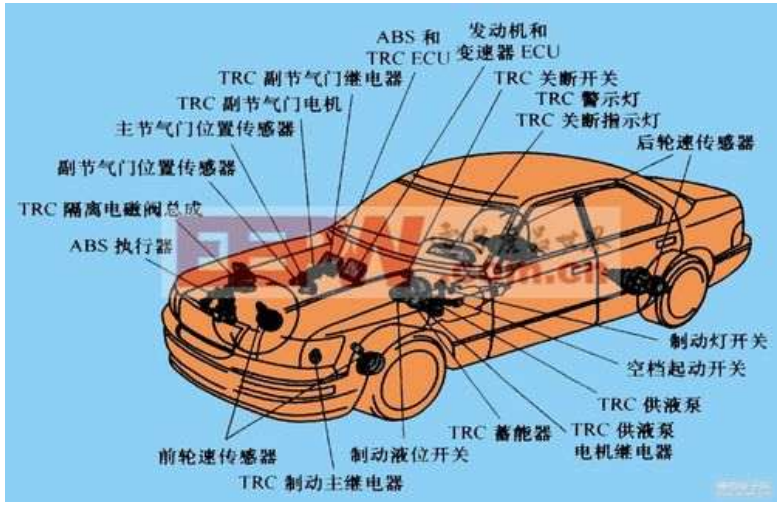
- The forming methods of parts show a trend of blooming.
- The development of computer science and technology provides powerful new technical support for the development of manufacturing technology.
- Accuracy and speed limits are constantly being broken through;
- Manufacturing technology has been deeply integrated with computer science and technology, electronic information technology, automation technology and management science.
- Digital intelligence has become the leading direction of manufacturing technology development;
- The rapid development of science and technology and society constantly puts forward new demands and challenges for the development of manufacturing technology.

two The development history of the relevant direction of this professional category-mechanical and electronic engineering development

1) Development history

- The digital computer was born in 1946 and entered the digital technology era.
- In 1952, the first CNC machine tool was born, forming the prototype of digital control of mechanical equipment.
- In the 1960s, a large number of CNC machine tools were popularized and applied, which demonstrated and laid the foundation for the numerical control of mechanical equipment, and then flourished and popularized.
- Mechatronics was defined in a supplement to the Japanese magazine Mechanical Design in 1971.
- In 1993, Bochum University of Applied Sciences established an independent mechanical and electronic engineering department.
- China began to pilot majors in 1980s, and began to set up master's and doctoral programs in 1990, Harbin Institute of Technology was the first in China.



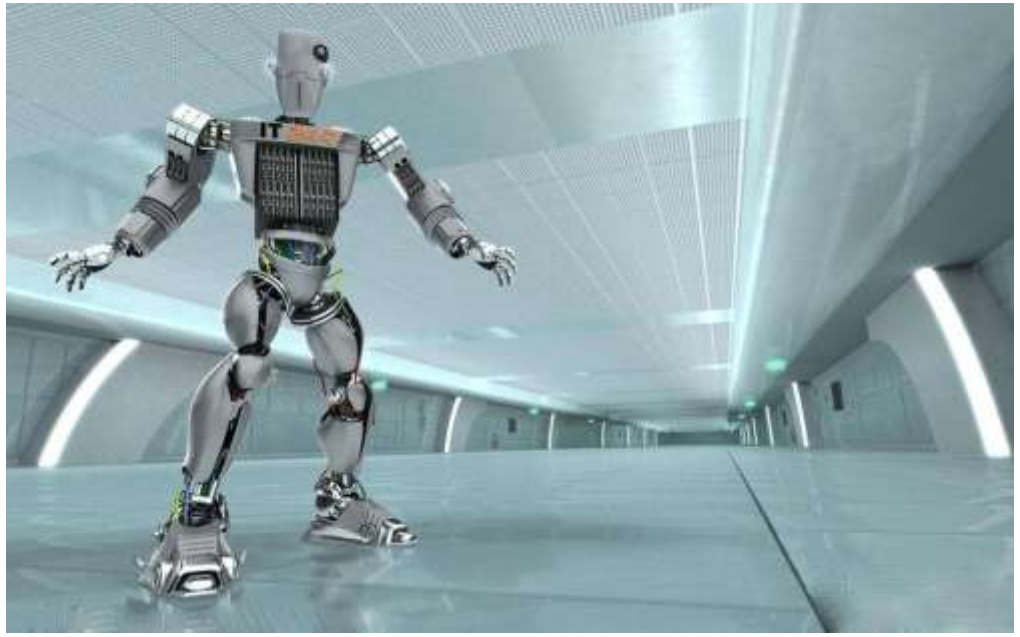


two The development history of the relevant direction of this professional category-mechanical and electronic engineering development

2) Connotation and related technology

- Mechanical and electronic engineering, commonly known as mechatronics, is a branch of mechanical engineering.
- It is a collection of mechanical, electronic, control, information, computer, artificial intelligence, management and many other theoretical systems. Its characteristics are huge knowledge structure, rich theory and wide application range. Mechanical and electronic courses are mainly composed of mechanical engineering, electronic engineering, computer science and control engineering courses.
- It involves theories, methods and technologies of mechanical design and manufacturing, electronic information, computer, automatic control, sensing technology, information processing, servo drive, artificial intelligence, system integration, etc.
- Robots are one of the typical examples.





two The development history of the relevant direction of this professional category-mechanical and electronic engineering development

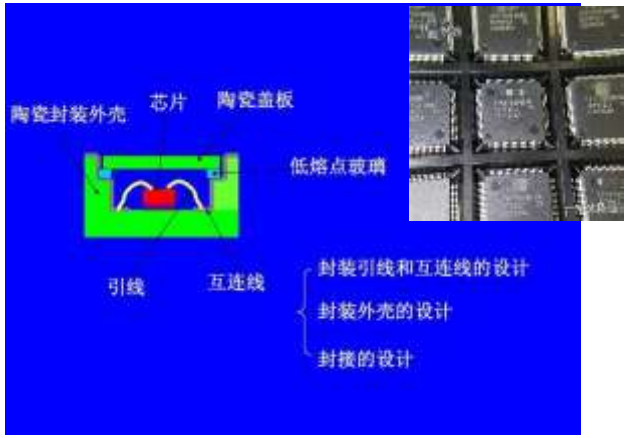
3) Brief summary

- The development of mechanical and electronic engineering accelerates the rapid development of digitalization and automation technology of electromechanical equipment, and makes the design and manufacture of electromechanical systems step into a new era.
- So far, almost all electromechanical products have used electromechanical integration technology.
- The development of mechatronics technology has laid a solid foundation for the application of artificial intelligence technology in electromechanical equipment and expanded a broad space.









two The development history of the relevant direction of this professional category-robot engineering development

1. Robot concept and research content

(1) Definition of robot:

Robot is an electromechanical device with the function of simulating human or animal. It is an automatic, position-controllable and programmable machine.

(2) Main research contents:

- robotics
- robot kinematics
- robot statics
- robot dynamics
- robot control technology
- Robotic perception technology
- Robot motion planning method
- robot programming language
- Robot intelligence technology
- Special materials for robots

two The development history of the relevant direction of this professional category-robot engineering development

2. Review of the development

1) The first stage of robotics development (before the birth of digital computers)





(1) Representative cases:

- During the Western Zhou Dynasty, actors who could sing and dance well.
- Late spring and autumn "three days do not fall" wooden birds.
- Moving Greek statues-automata of the 2nd century BC.
- Jili Drum Car in Han Dynasty of China.
- Wooden oxen and horses in the three kingdoms period.
- .....
- In 1920, Czech writer Karel. Capek's "Rosam's Omnipotent Robot" uses Robot.





The development history of the relevant direction of this professional category-robot engineering development

1) The first stage of robotics development (before the birth of digital computers)

(2) Main contributions and summary:

- **Invented ingenious mechanisms that mimic certain human or animal movements.**
- **The power used basically comes from human, stored power, natural power or steam power.**
- **The primary use is for entertainment, material handling, or to satisfy a curiosity.**
- **The initial concept of Robot was put forward, and the English term has been used until now.**

The development history of the relevant direction of this professional category-robot engineering development

2) The second stage of robot engineering development (after the birth of digital computers)

(1) Representative work:

- The development of digital computers and CNC machine tools has further opened up people's imagination of robots and laid the foundation for the development of robots.
- The Argonne Institute of the United States developed the teleoperation manipulator in 1947 and the master-slave manipulator in 1948.
- 1965 MIT Robotic system with visual perception.
- 1967 Japan's first robotics conference.
- 1970 First International Conference on Industrial Robotics, USA.
- 1973 Cincinnati Company develops microcomputer-controlled industrial robot
- After the 1980s, robots entered a period of rapid development, and all kinds of robots flourished and formed a system.

The development history of the relevant direction of this professional category-robot engineering development

2) The second stage of robot engineering development (after the birth of digital computers)

(1) Representative work:

- China began research in the 1970s, focusing on robot mechanics and robot kinematics.
- China's 1980s robot prototype development, Harbin Institute of Technology's first demonstration robot, the first arc welding robot.
- China developed practical robots in the 1990s, began industrial application demonstration, and entered the commercialization stage.







The development history of the relevant direction of this professional category-robot engineering development

2) The second stage of robot engineering development (after the birth of digital computers)

(2) Main contributions and summary:

- The knowledge system of robotics has been formed and the industrialization of robots has been realized.
- Power is mainly electric power, mobile power is widely used.
- The application of artificial intelligence in robots has been realized and developed vigorously. The future prospects are broad.
- The diversified development of robots has been realized, and robots of different types and different purposes are widely used.









Thank you!

